Free-fall lab

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Estimate g from measured motion and discuss measurement limits.

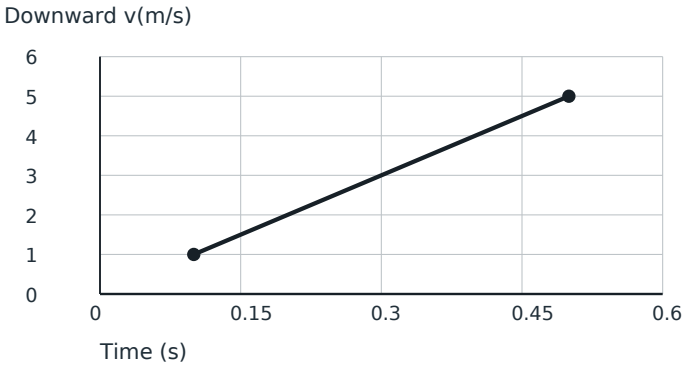
Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 2, Day 5.*
3 N, attractive toward the other mass.

02. Worked example: explain the model
Points 8, 10, 12 with mean 10; timing scale or calibration error could shift all readings.

03. Read and interpret the model
(5-1)/(0.5-0.1)=10 m/s² downward.

Model / diagram check



04. Apply the physics
Mean 10 m/s²; range 9-11. Variation helps estimate uncertainty and review technique.

05. Explain your reasoning
Controlled release/setup; height/time orv data; pushed release changes initial condition. Preserve flagged trial and explain exclusion if justified.

06. Review: connect an earlier idea *Review: Unit 1, Day 18.*
14 N left; 18-4-14=0 N.

07. Exit check
a = (6 - 4)/0.4 = 5 m/s² downward. This measurement is below the ideal g ≈ 10 m/s²; it should not be silently replaced by the expected value.

08. STRETCH: test the model
Both means 10; first group smaller spread and better repeatability. Accuracy needs a reference and systematic-error review.

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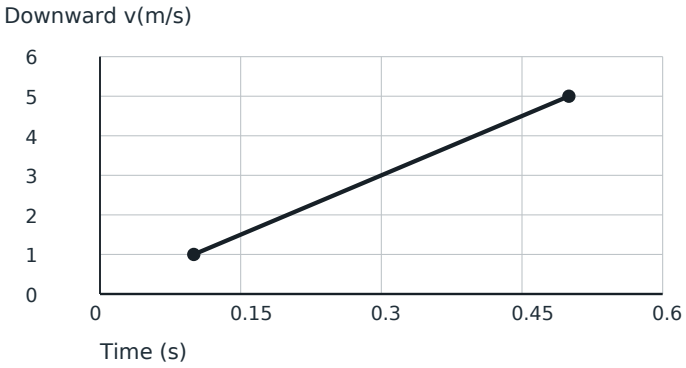
Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 2, Day 5.*
$F \approx 2.401 \times 10^{-5}$ N, attractive. The incorrect numerical answer is 5 times too large when SI values are used.

02. Worked example: explain the model
Points 8, 10, 12 with mean 10; timing scale or calibration error could shift all readings.

03. Read and interpret the model
(5-1)/(0.5-0.1)=10 m/s² downward.

Model / diagram check



04. Apply the physics
Mean 9.9; half-range .3 m/s²; deviation≈1.02%. Half-range is a simple spread estimate, not automatically a confidence interval.

05. Explain your reasoning
Controlled release/setup; height/time orv data; pushed release changes initial condition. Preserve flagged trial and explain exclusion if justified.

06. Review: connect an earlier idea *Review: Unit 1, Day 18.*
At rest T=39.2 N. Accelerating: T=m(g+a)=51.2 N.

07. Exit check
9.8 m/s²; multiple points reduce sensitivity to individual noise and reveal trend.

08. STRETCH: test the model
Mean 9.8, half-range 0.4 m/s², deviation≈0.102%. No; a shared calibration bias can give consistent but inaccurate values.